

I Create, Therefore I Agree: Exploring the Effect of AI Anthropomorphism on Human Decision-Making

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ABSTRACT

Artificial intelligence (AI) has been widely adopted daily, and much academic effort has been directed to investigate the influencing factors of human acceptance and collaboration with it. Anthropomorphism can effectively foster positive evaluations of inanimate objects, and AI is no exception. The current study investigates how naming and personalizing an AI prototype's appearance affect the human decision in a recipe ingredient recommendation system and finds that people are more likely to accept AI's recommendation if they could name and customize its appearance than if they did not have the options.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

artificial intelligence, decision-making, anthropomorphism, naming

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1 BACKGROUND

Artificial intelligence (AI) has been increasingly integrated into various aspects of daily life, such as in decision-making and recommendation systems [1]. The more recent ChatGPT and other generative AI services have also created another hype across different populations [27]. Many individuals and industries have benefited from AI technologies that aid their task execution [8]. In interactions, humans develop different attitudes and perceptions toward AI technologies [20]. These attitudinal shifts may affect their acceptance of and collaboration with teammates [8].

Despite AI's vast and fast integration into the workforce, research on what factors may influence people's attitudes toward AI teammates is still just budding. Current literature suggests that factors such as an AI's personality [20], the perceived competence

and warmth [8], and its influence on the team's shared resources [6] may all play a role in affecting people's attitudes and acceptance of, as well as collaboration or interaction with AI. Additionally, people's mental models of AI agents, the representations formed in their minds [17], may also impact their expectations and subsequent usage of the agents. It is likely that they treat them unintentionally as social actors rather than machines [23] and sometimes build expectations based on the name, which can deviate from what the agent is actually capable of, and therefore lead to misalignment and dissatisfaction when using the agent [17].

There are different ways to foster the relationship between humans and AIs; anthropomorphism is a relatively common method [13]. Anthropomorphism refers to assigning human-like traits, mental states, and behavioral patterns to non-human entities [24]; naming and customization of the appearance can be seen as two types of such assignment.

Naming has been a common symbolic behavior throughout human history, and it is socially important for interpersonal relationships across cultures [11]. People give names to newborns, pets, and even objects and project identity, personality, and expectation throughout this process [25]. Naming and, in some cases, nicknaming create or reinforce the socio-emotional bonds in interpersonal relationships [18, 25].

This effect is also carried over in human-object relationships, as naming one's belongings, such as cars and computers, is prevalent in many societies [22]. The naming of objects entails several interesting consequences, including affecting one's first impression and subsequent attitudes toward new technologies [2, 16, 29] and increasing the positive evaluation and purchasing intention of products [12, 15]. Having people name objects on their own could further the psychological ownership of the products in marketing [25] and can sometimes be used to convey their attitudes and thus become a part of their own identity [28].

In addition to naming, giving something an anthropomorphic appearance has been found effective in bringing positive evaluation in social robotics (e.g., [10]) and provides rich opportunities for human-robot interaction [30] as well as interaction with AI-enabled chatbots [5]. However, some scholars have also warned us of the potential risks of anthropomorphizing agents, as such a tendency may bring harm to hybrid teamwork by exposing military personnel to lethal risks in a military setting [4], or introduce cognitive biases in favor of the machine/agent in judgment [21].

In spite of the prevalence of the idea of anthropomorphizing agents and the historically significant role naming has played in shaping social relationships, much still remains unknown about the role of naming and anthropomorphizing an agent's appearance in the context of AI, a gap the current study aims to explore. Therefore

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the guiding research question for the current work is: *How does anthropomorphism through naming and customizing AI's appearance affect users' decision-making results?*

We found that people were more likely to accept AI's recommendation if they could name and customize its appearance than if they did not have the options. This work contributes to our understanding of how anthropomorphism affects human decision-making in the context of collaboration with AI and opens up more research opportunities to foster a positive relationship between AI and human users.

2 METHOD

2.1 Procedure

In this study, we used a between-participant design in which participants were randomly assigned to one of the four conditions after they read and signed the online consent form. Participants were then directed to an experiment website, where the AI's function was introduced through texts and a sketch of the AI.

Once introduced to the AI, participants were given an *ingredient replacement task* in which they had to choose a substitute for an ingredient that was unavailable while cooking a recipe. The AI suggested seven different ingredients in the order of decreasing scores between 0 and 1, with higher scores indicating stronger recommendations. Participants were required to select one ingredient from the list and provide a brief explanation for their choice.

After finishing the task, participants were directed to an online survey where they answered questions about their demographic information and general attitudes toward AI. In the end, they submitted their worker's IDs to the Amazon Mechanical Turk platform and received 1.25 USD for their participation. This task can be completed in 8-10 minutes. That is, each worker can get at least 7.5 USD per hour, above the minimum wage in the United State [26].

2.2 Materials

2.2.1 Conditions, task, and interface. The four conditions were (1) **self-naming and customizing**, where participants were asked to give a name to the AI and choose one of the hat options (i.e., a cooking hat, a baseball cup, and none), (2) **chef with a chef hat**, where the participants were told the AI's name was "Chef" and the sketch of the AI was wearing a chef's hat, (3) **Chris with a baseball cap**, where the participants were told the AI's name was "Chris" and the sketch was wearing a baseball cap, neither of which insinuate the AI's professional identity, and (4) **baseline**, where the AI has a generic name "AI" with no hats in the sketch (see Figure 1).

In this work, we aim to create a domain-approachable task for the general public by designing an ingredient replacement scenario. This scenario represents a common everyday decision-making situation that individuals may frequently encounter in their daily lives. Prior research has also used similar ingredient replacement tasks to explore human trust and over-reliance in AI-assisted decision-making [3]. However, unlike previous work, we opted not to use an imaginary scenario. Instead, we developed a preliminary recommendation system based on an open-source dataset. To ensure practicality and real-world relevance, we used data from Recipes5k, a dataset containing 4,826 distinct recipes. Each recipe is composed

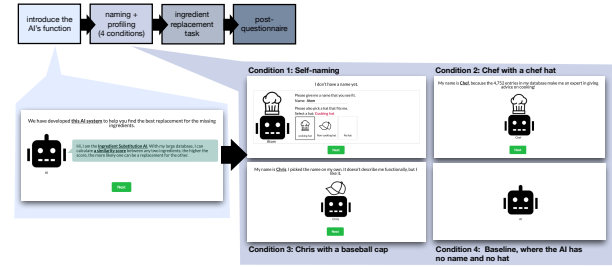


Figure 1: We first introduced the AI's function with texts and then directed the user to one of four conditions

of an image along with a list of ingredients for the purpose of ingredient recognition. The dataset includes a total of 3,213 unique ingredients, with an average of ten ingredients per recipe¹.

Our preliminary recommendation system operates on the basis of ingredients similarity. To calculate the similarity between ingredients, we employed the *Jaccard similarity*, which is a basic method for measuring the similarity between two entities. We calculated the asymmetric relative co-occurrence between ingredients, assuming that two ingredients are more likely to be similar if they frequently appear or be used in the same recipe. Based on this premise, we identified the top seven substitute ingredients with the highest similarity scores for each target ingredient. In the end, we selected one recipe with a specific target ingredient that our system provided the most relevant suggestion for its substitution. While the use of asymmetric relative co-occurrence as a measure of ingredient similarity may not offer highly accurate recommendations for the ingredient replacement task, this research does not delve into the impact of recommendation accuracy on user perception and decision-making. Future studies can explore the integration of more precise or state-of-the-art recommendation techniques for further enhancements.

The task interface is composed of two parts: one is the recipe with AI's suggestion, and the other is the decision-making task. The first part of the task is to show AI's profile and its suggestion. We used a profile image with a name to represent the AI. The second part of the task is to ask participants to perform a decision-making task. They are asked to choose a substituting ingredient and provide a brief explanation for their choice (see Figure 2).

2.2.2 Measures. To measure the participants' perception of the AI prototype, we adopted Ashoori and Weisz's [1] scale that evaluates the different facets of trust for AI with its sub-scales for trustworthiness, reliability, technical competence, and personal attachment. Participants indicated their level of agreement on a 7-point Likert scale (1 = extremely disagree, and 7 = extremely agree).

To measure the participants' general attitudes toward AI technologies, we used the questions from Gillespie et al.'s five-country study on trust in AI [7] and specifically asked about participants' support for AI systems ("How much do you support or oppose the development and use of AI?") and their AI acceptance ("In thinking about AI, which of the following best represents your view?"). Users indicated their opinions on a 7-point Likert scale (1 = strongly

¹<http://www.ub.edu/cvub/recipes5k/>

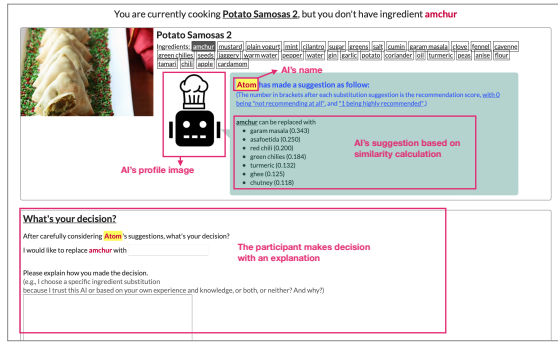


Figure 2: The ingredient replacement task interface consists of two parts: the top part is to show the recipe with AI's suggestion and the bottom part is the area of collecting the user's decision.

oppose and 7 = strongly support) and a 5-point Likert scale (1 = I reject AI, 2 = I tolerate AI, 3 = I accept AI, 4 = I approve AI, and 5 = I embrace AI), respectively.

We also measured participants' perceived cognitive workload using the NASA TLX [9] as well as participants' demographic information, including gender, age, race, religion, and highest degree completed.

2.3 Participants

We recruited 80 participants on Amazon Mechanical Turk (MTurk), with qualification requirements of (1) HIT approval rate over 95%, (2) over 500 HITs have been approved, and (3) the location is either in UK or US, and randomly assigned them into one of the four conditions. After data collection was complete, we went through the responses to filter out irrelevant ones and ended up discarding 39 participants' data as they either did not provide valid answers to our question or failed to pass the attention check question embedded in the survey. The final sample included 41 participants.

Participants' age ranged between 24 and 64 years old ($M = 35.80$, $SD = 9.82$). Twenty-one were male, and twenty were female. Thirty-five participants identified as Caucasians, 3 African American, and 3 Asian or Pacific Islanders. Thirty-three held a bachelor's degree or higher.

3 RESULTS

We ran a Chi-square test for homogeneity and post hoc pairwise comparisons for the choices of new ingredients. As shown in Figure 3, results indicated that there was a significant difference ($\chi^2 = 4.70$, $df = 1$, $p = .03$) between Condition 1 and Condition 2. Specifically, participants whose AI was a chef wearing a chef hat were less likely to accept the AI's recommendation than participants who named their AIs and chose the appearance on their own.

Additionally, there was also a significant difference between Condition 1 and Condition 4 ($\chi^2 = 3.88$, $df = 1$, $p < .05$). In other words, participants who chose names and appearances for their AIs were more likely to accept their suggestions than when they did not have any choice.

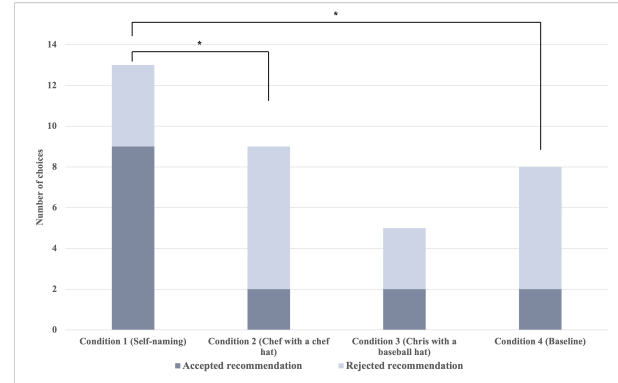


Figure 3: Decision of accepting/rejecting AI recommendation across the four conditions

We found no relationship between participants' general attitudes toward AI technologies or support and acceptance of AI and their task choices. Neither did their perceived task loading vary across conditions.

4 DISCUSSION

Our results suggested that participants who were given the option to name their AI were more likely to accept the recommended ingredient than those who did not have the option. Although this corroborates the previous finding that naming an inanimate object increases the positive evaluation of it (e.g., [12]), we did not find any relationship between a higher acceptance rate of recommendations and the perceived trustworthiness of the AI, leading us to conjecture that the positive evaluation, or in our case, the higher acceptance rate of AI's suggestion, is not achieved through a recognition of AI's competence, but through something else, such as a psychological sense of ownership of the AI, and a subsequent liking for it.

The Elaboration Likelihood Model posits that multiple factors may be at play in decision-making, such as our likeness for someone, celebrity endorsement, and perceived intimacy with a persuader [14, 19]. Despite being identified as part of the "peripheral route" rather than the "central" and more cognitively demanding one, their influence on human behaviors has been widely confirmed in various contexts. Our study suggests that even in collaboration with AI, going through the peripheral route and being affected by cues such as psychological ownership or closer perceived intimacy may be at play. This finding also harks back to the potential cognitive bias that anthropomorphism may introduce when human users interact with such an agent [21], in that people's judgment may indeed be clouded when they get the opportunity to name and claim ownership of an agent. However, we did not measure the sense of ownership in the current study, which could be an interesting future direction to explore further.

In terms of the broader field of human-robot interaction or human-agent interaction, our work preliminarily explored how giving a name and personifying an agent might affect the way people interpret the results. Although our way of operationalizing the concept of anthropomorphism was relatively simple in the experiment, it has opened further space to explore in future studies in

terms of the impact of different ways of anthropomorphism on the collaboration with agents, as well as in other contexts, such as one where the agents/robots serve for social/emotional purposes (e.g., accompanying the seniors, socializing with young children).

We recognize that there are two major limitations of the current study. The first is the small and unbalanced sample we recruited on MTurk. We had to filter out almost half of the data due to the unqualified answers. Given a bigger and more balanced sample, the results might be different. The second limitation is that the task we chose was not particularly high stake and, therefore, might not be able to prompt the participants to be more elaborate on their decision-making process. In other words, they might not have enough motivation to consider the recommendations given by AI in a more rigorous manner. Future studies should address the potential impact of the stakes of the task.

5 CONCLUSION

With the fast and vast integration of artificial intelligence in our daily life, increasing academic effort has been directed to investigate the influencing factors of human acceptance and collaboration with it. The current study investigates how naming and personalizing the appearance, both methods to anthropomorphize an AI prototype, affect the human decision in a recipe ingredient recommendation system. We found that people are more likely to accept AI's recommendation if they could name and customize its appearance than if they did not have the options. This opens up more research opportunities in fostering a positive relationship between AI and human users.

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